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Hamburg, 10 March 2026

Prof. Dr. Matthias Schwab & Dr. Elke Schaeffeler
Dr. Margarete Fischer-Bosch Institute of Clinical Pharmacology
Robert-Bosch-Krankenhaus
Auerbachstraße 112
70376 Stuttgart

Post Doctoral Research Scientist - Bioinformatics (m/f/d)

Dear Prof. Dr. Schwab and Dr. Schaeffeler,

My experience refactoring production-grade medical data pipelines has taught me how crucial robust computational infrastructure is for clinical research. The opportunity to elucidate the tumor microenvironment by integrating spatial transcriptomics and scRNA-seq data at your institute directly matches this expertise. My transition from a Ph.D. in computational systems modeling to bioinformatics pipeline engineering makes me a highly suitable candidate to develop the novel computational solutions your clinical research demands.

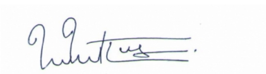
While my academic background is in systems modeling, I have spent the last year intensively retraining in modern bioinformatics, transcriptomics, and machine learning. At my current internship with HealthTwiSt GmbH, I am solely responsible for refactoring a production medical data pipeline (in R/tidyverse) for Germany's interventional radiology quality registry. Working with 143,000 patient records and 1,920 variables, I externalized hardcoded logic into configuration files, verified byte-level reproducibility, and reduced the 1,834-line codebase by 26%—establishing the robust biostatistics foundation required for integrating clinical phenotypes with multi-omics data.

Your project's need to bridge single-cell datasets with clinical outcomes requires deep expertise in both programming and data integration, which are my core strengths:

- **Bioinformatics & Transcriptomics:** Reproducible, containerized workflows (Nextflow/DSL2) for multi-omic data; strong statistical modeling for high-dimensional datasets like spatial transcriptomics and scRNA-seq.
- **R and Python:** Clinical data workflows (Tidymodels, Bioconductor, tidyverse) and data analysis pipelines (Pandas, scikit-learn, JAX) for prototyping new computational solutions.
- **Large-Scale Data & Oncology:** Processed terabytes of experimental data on HPC clusters; eager to apply expertise in coupled systems to understanding metastasis and the tumor microenvironment in renal cell carcinoma.

I am deeply motivated by the Institute of Clinical Pharmacology's mission to bridge omics data with clinical phenotypes. My robust mathematical background and modern data engineering skills will allow me to immediately contribute to your interactive team of bioinformaticians, clinicians, and biologists. I am available from May 2026 and welcome the opportunity to discuss my candidacy in a personal interview.

Sincerely



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Enclosures: Curriculum Vitae, Degrees and Certificates